Through Zero and Beyond – One Step at a Time

Questions

1. Fill in the blanks for the missing numbers on each scale.

a)

–3 –2 ___ 0 1 ___ 3

b)

___ –1 0 1 ___ 3 4

c)

–6 ___ –4 –3 ___ –1

d)

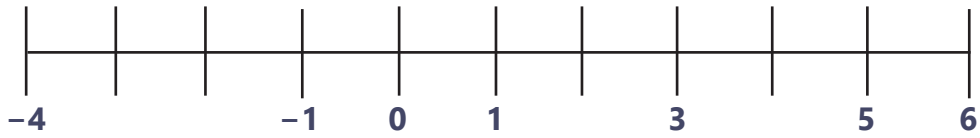
___ –9 –8 ___ –6 ___ –4

2. Complete these number lines.

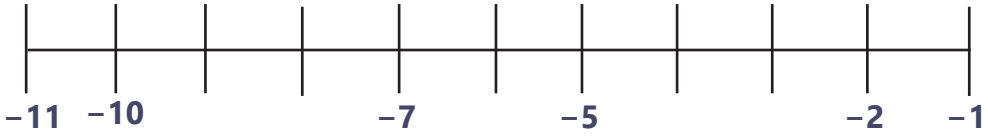
a)



b)



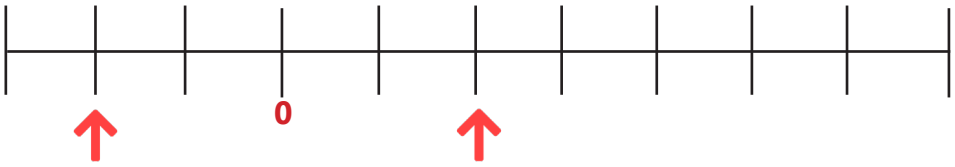
c)



3. If you start at **–5** and take **6 steps** forward in **1s,** where do you land? Show your working.

© Pencilmaths 2025

4. The arrows are pointing to numbers on a number line.



What numbers are the arrows pointing to?

5. Rani and Arjun are counting backwards through zero.

Rani: 3, 2, 1, 0, –2, –3, –4

Arjun: 5, 4, 3, 2, 1, –1, –2

a) What mistake has Rani made?

b) What mistake has Arjun made?

6. Write the next four numbers in each sequence.

a) 4, 3, 2, 1, ___, ___, ___, ___

b) –4, –3, –2, –1, ___, ___, ___, ___

c) –10, –11, –12, –13, ___, ___, ___, ___

d) 2, 1, 0, –1, ___, ___, ___, ___

7. Here are Thursday's temperatures in four different cities:

City	Temperature
Berlin	**3°C**
Helsinki	**–5°C**
Madrid	**5°C**
Warsaw	**–2°C**

a) On Saturday, Berlin is **6°C** colder than on Thursday. What is the temperature?

b) On Saturday, Helsinki is **4°C** warmer than on Thursday.

c) A city has a temperature exactly between Madrid and Warsaw. What is the temperature in that city?

Pencil Maths

8. A diver is floating at the surface (0 m). He dives to –4 m, then descends another 5 m.

a) What mistake might someone make if they think the diver is at –8 m?

b) What depth is the diver now at?

9. Leo says, "If I count from –3 up to 2, I will say five numbers."

Is Leo correct? Explain how you know.

10. Ravi says, "There are the same number of negative numbers as positive numbers between –3 and 3." Do you agree with Ravi? Why or why not?

Pencil Maths